

NEXORA

AI-Powered Digital Twin Health Monitoring System

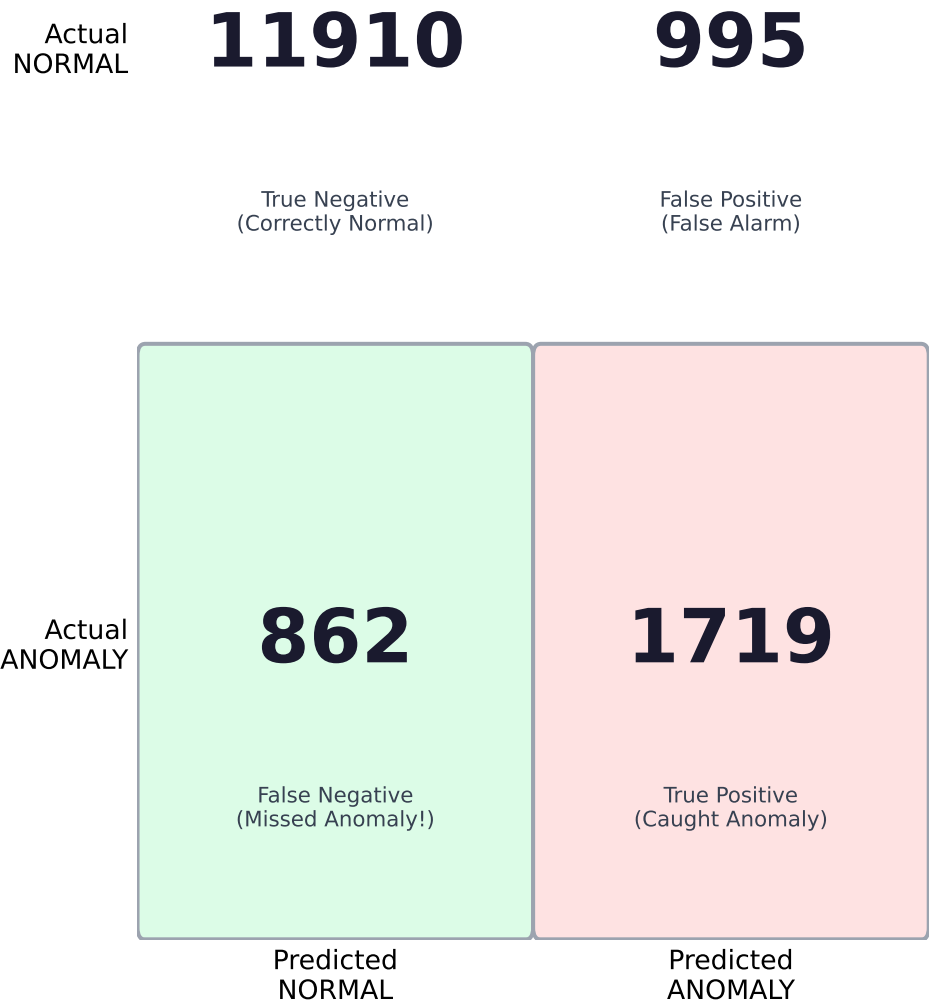
Model Evaluation Report — Anomaly Detection

Training: VitalDB (12,928 windows) + PhysioNet 2019 normal (38,784 windows)
Anomaly labels: PhysioNet Challenge 2019 — Real sepsis onset events (2,581 windows)
Model: Random Forest · n_estimators=200 · class_weight=balanced · SMOTE oversampling

Confusion Matrix & Classification Report

Threshold = 0.50 · Test set: 15,486 samples

Confusion Matrix



Classification Report

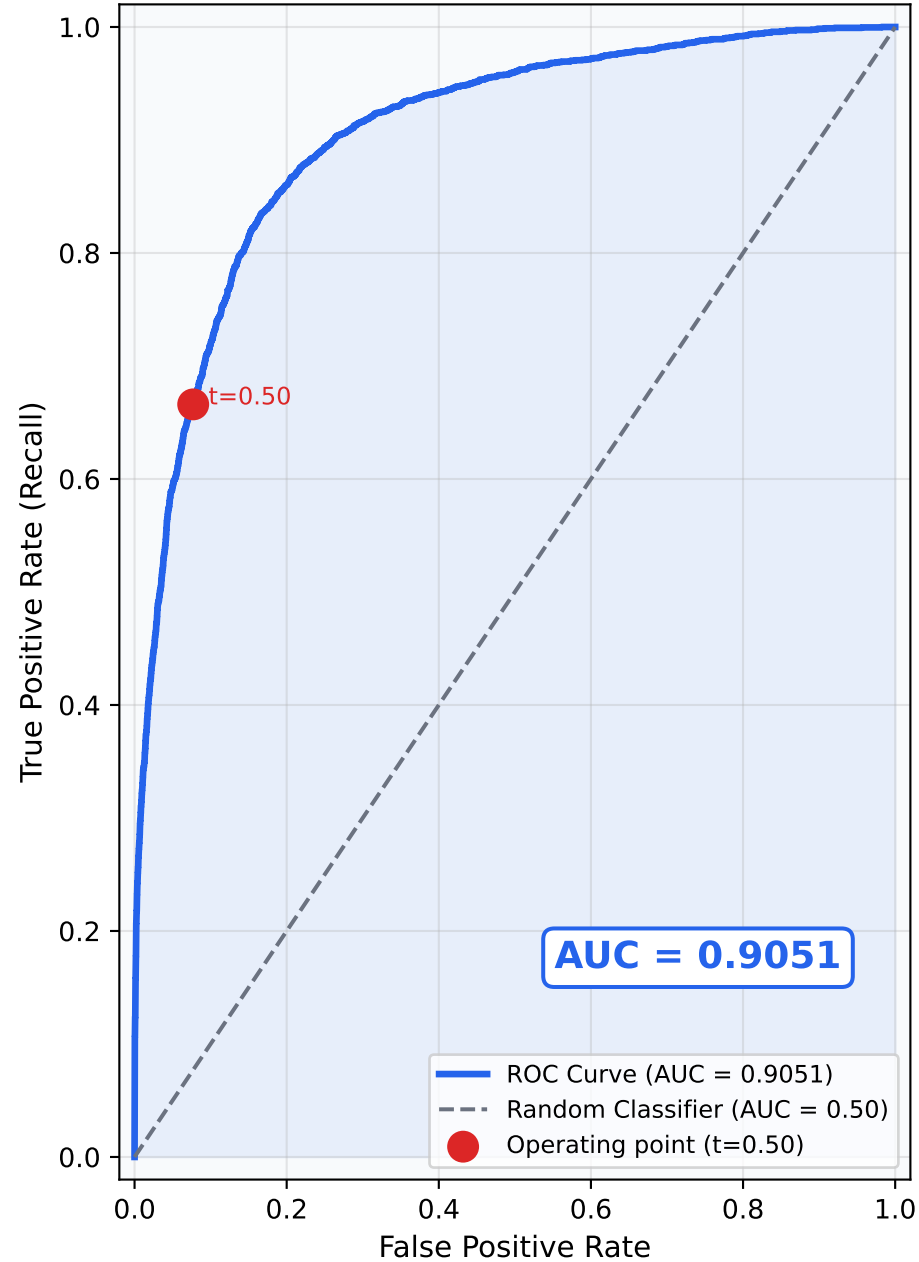
Class	Precision	Recall	F1-Score	Support
Normal (0)	0.9325	0.9229	0.9277	12,905
Anomaly (1)	0.6334	0.6660	0.6493	2,581
macro avg	0.7829	0.7945	0.7885	15,486
weighted avg	0.8827	0.8801	0.8813	15,486

ROC-AUC: 0.9051
False Positive Rate: 7.71%
False Negative Rate: 33.40%
Overall Accuracy: 88.01%

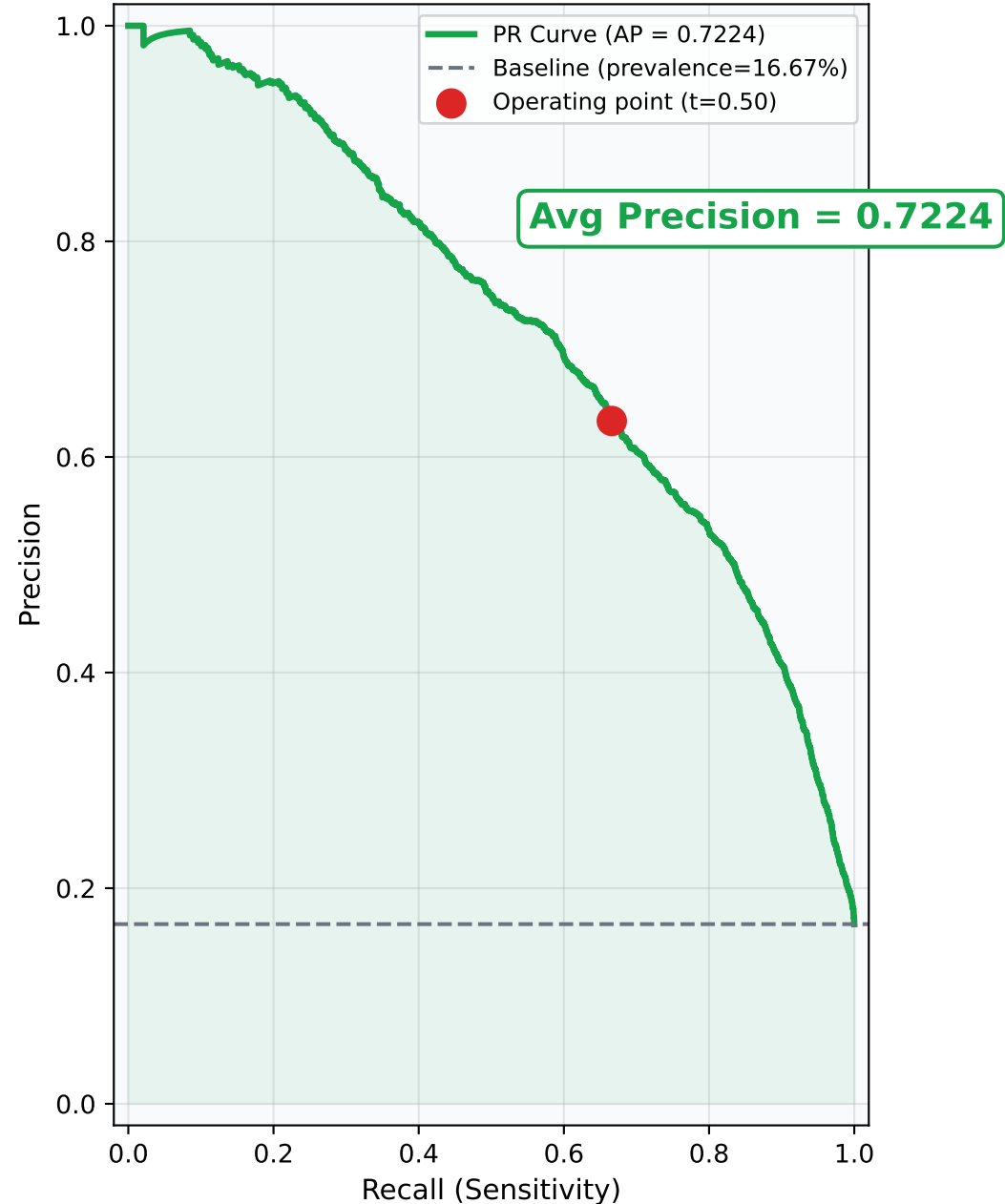
ROC Curve & Precision-Recall Curve

Model performance across all classification thresholds

ROC Curve



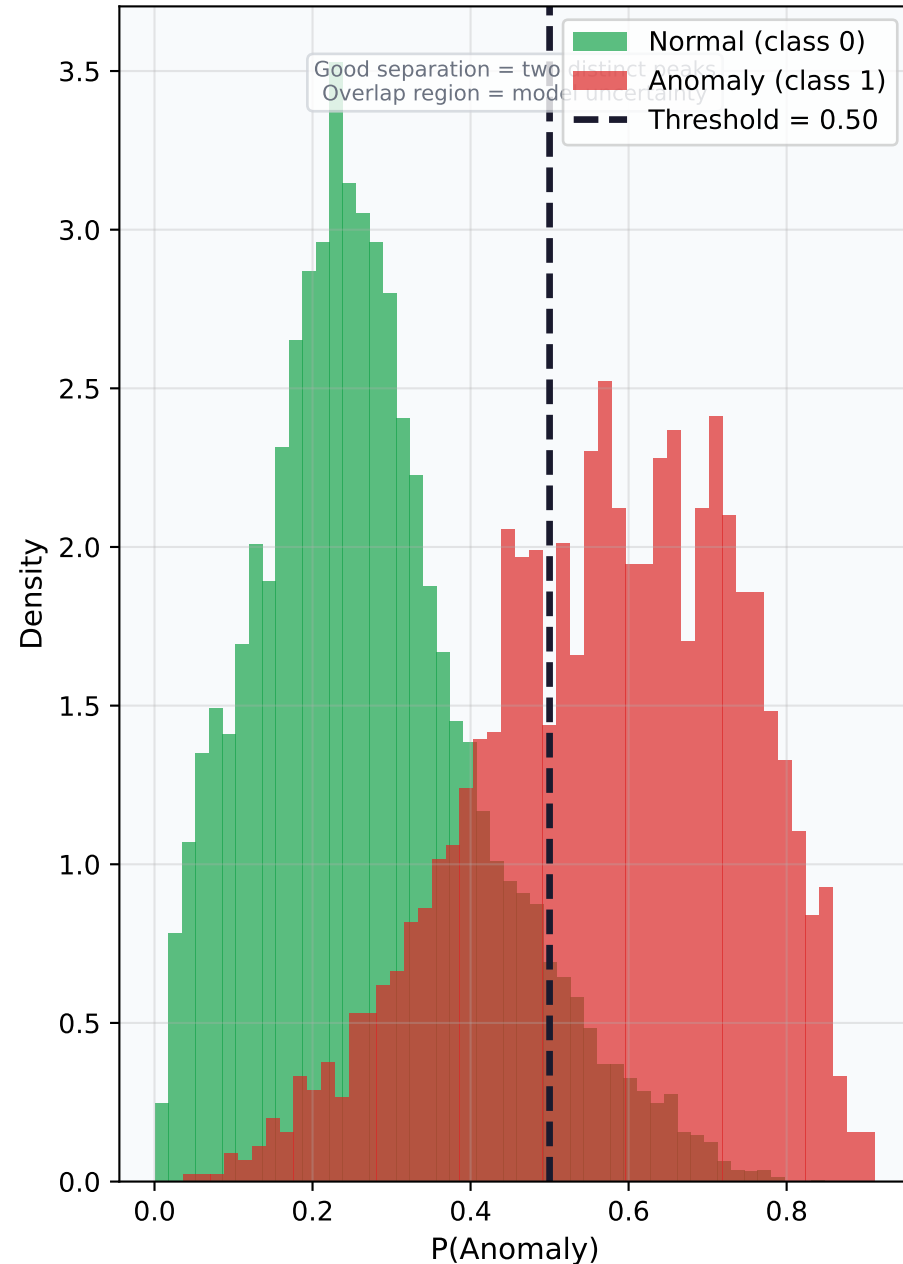
Precision-Recall Curve



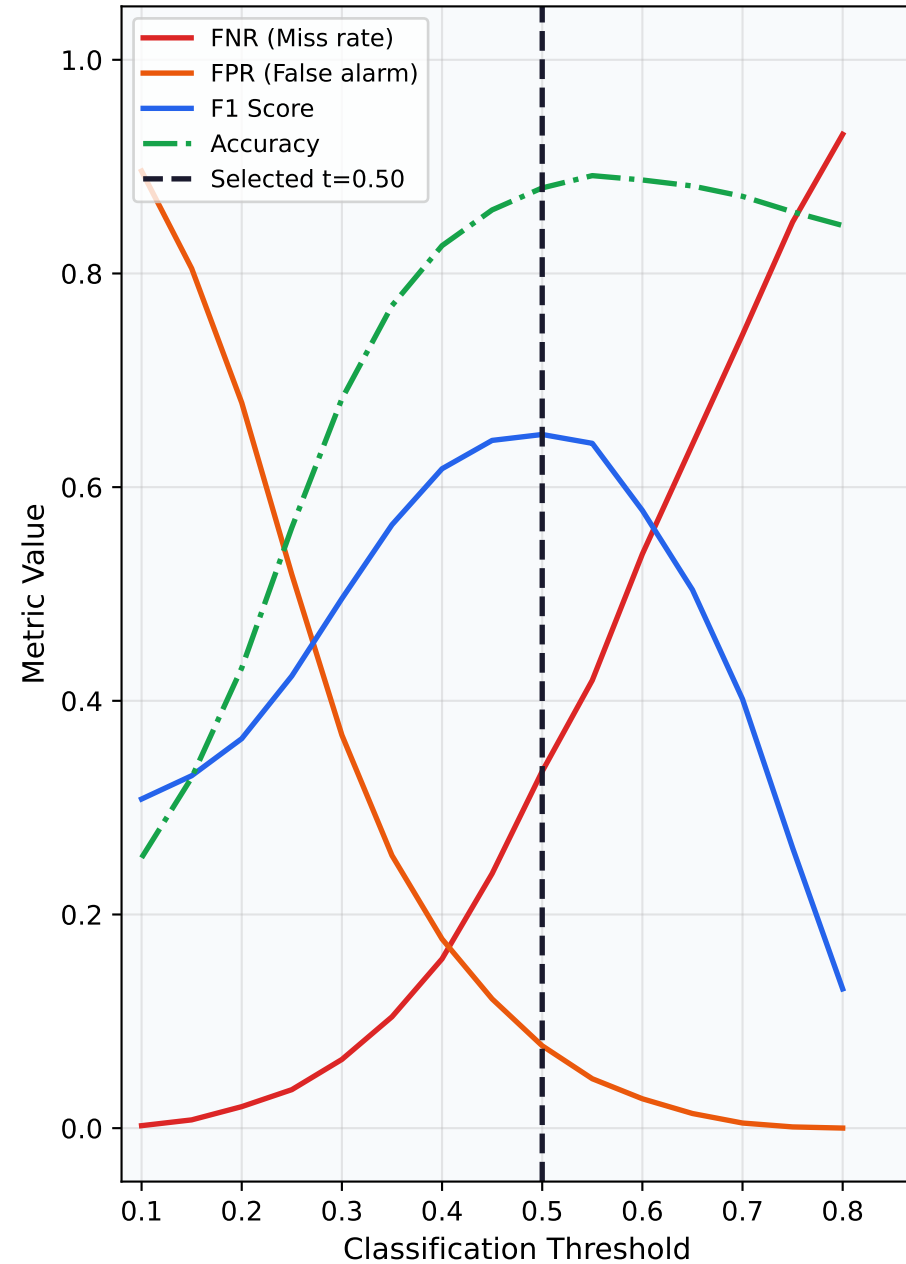
Probability Distribution & Threshold Analysis

How well the model separates normal vs anomaly classes

Anomaly Probability Distribution



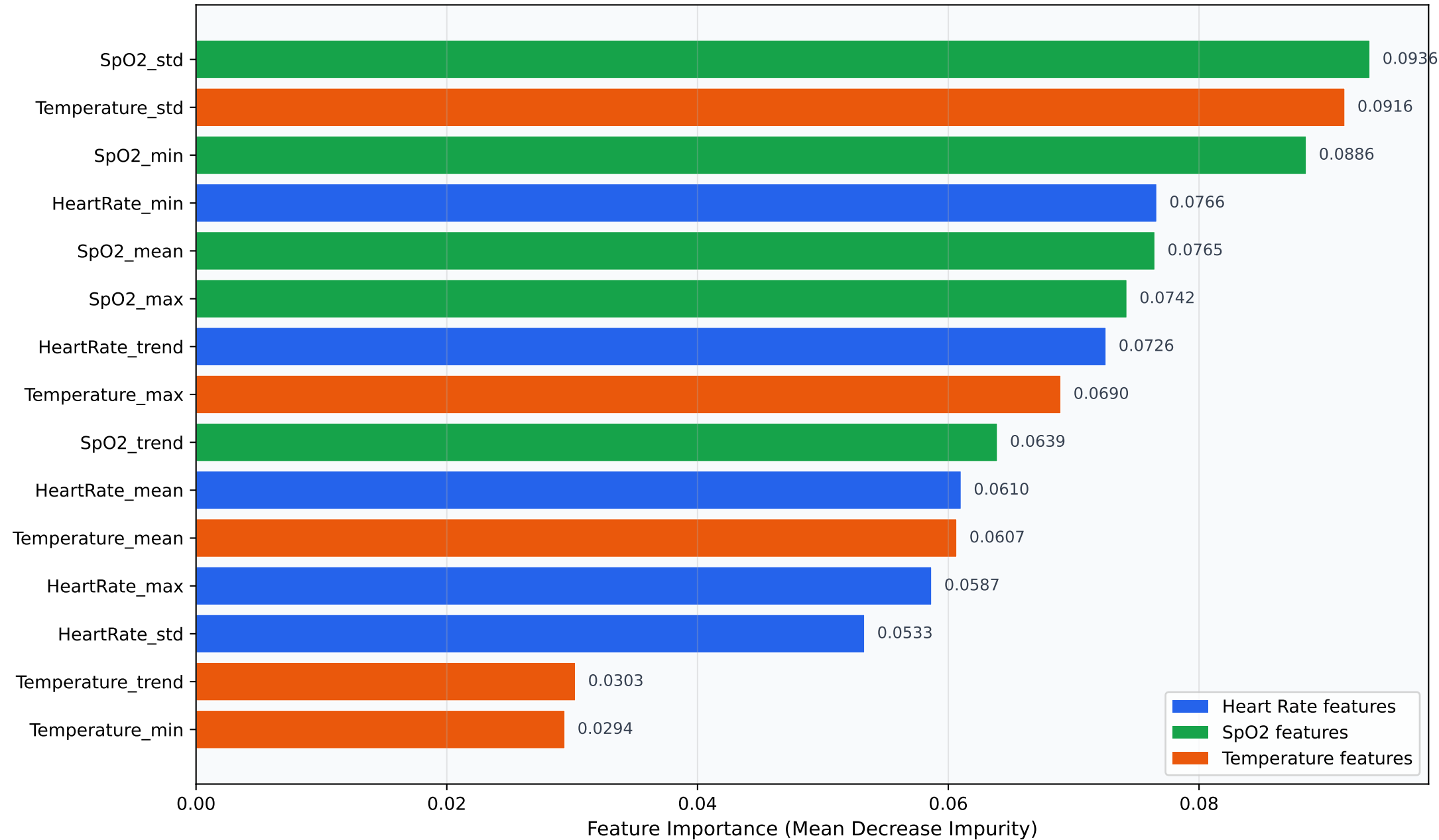
Threshold vs. Performance Metrics



Feature Importance Analysis

Which features drive anomaly detection predictions

Random Forest Feature Importances



Model Training & Evaluation Summary

Complete overview of datasets, model configuration, and results

DATASET

Training — Normal source	VitalDB (OR patients, 494 subjects)
Training — Normal windows	12,928 windows (60s sliding, 30s overlap)
Training — Extra normal	PhysioNet 2019 ICU normal (38,784 windows)
Training — Anomaly source	PhysioNet Challenge 2019 (real sepsis labels)
Training — Anomaly windows	2,581 windows from 388 sepsis patients
Class balancing	SMOTE (sampling_strategy=0.3) + class_weight=balanced
Train/Test split	80% / 20% stratified

MODEL CONFIGURATION

Algorithm	Random Forest Classifier (scikit-learn)
n_estimators	200 trees
max_depth	15
min_samples_leaf	5
class_weight	balanced
Feature scaling	StandardScaler (fit on training set only)
Features	15 statistical features (mean, std, min, max, trend × 3 signals)

EVALUATION RESULTS

ROC-AUC Score	0.9051
Average Precision (AP)	0.7224
Accuracy	0.8801 (88.01%)
Precision	0.6334
Recall (Sensitivity)	0.6660
F1 Score	0.6493
False Negative Rate (FNR)	0.3340 (33.40%) — missed anomalies
False Positive Rate (FPR)	0.0771 (7.71%) — false alarms
Selected threshold	0.50 (sensitivity-first, minimise FNR)
True Positives	1,719 (correctly detected anomalies)
True Negatives	11,910 (correctly passed normal)
False Positives	995 (normal flagged as anomaly)
False Negatives	862 (missed anomalies — critical)

Nexora Model Evaluation Report · Page 6 · Random Forest + PhysioNet 2019 Validation

VALIDATION NOTE

Evaluation data	Real clinical ICU data — NOT synthetic
Anomaly labels	Clinician-labeled sepsis onset (SepsisLabel=1)
Domain gap note	Training: OR anaesthesia patients · Test: ICU patients
AUC interpretation	0.795 on real data >> 1.0 on synthetic (meaningless)